



Unleash your potential

Module Presenter's Manual

for

Web Framework for Python (Django)

***Effective From: June 2024
Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	June 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Learn the basics of Django, Templates, forms, Filters, and Serialization.
- Learn MVC pattern and data handling operations.
- Learn Data Serialization
- Work with Django's REST Framework

2. Information on Session Allocation

Module	Online Hours
Web Framework for Python (Django)	24

Throughout this Presenter's Manual, the module *Web Framework for Python (Django)* will be referred to as **DJANG**.

3. Session Deliverables

To aid the teaching process, following are the deliverables

(Faculties can access eBooks and other components in Onlinevarsity through Aptrack.)

Resources available on Onlinevarsity:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A Session is of 2 hours duration.

➤ Week-Wise Schedule

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 DJANG – TL1	Session 2 DJANG – TL2	Session 3 DJANG – TL3	Session 4 DJANG – TL4
2	Session 5 DJANG – TL5	Session 6 DJANG – TL6	Session 7 DJANG – TL7	Session 8 DJANG – TL8
3	Session 9 DJANG – TL9	Session 10 DJANG – TL10	Session 11 DJANG – TL11	Session 12 DJANG – TL12

DJANG: Web Framework for Python (Django)

TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	DJANG – TL1	All the topics as listed below from Session 1 and Session 2 of <i>Django Framework for Python</i> book should be covered in this session. <u>Session 1 – Introduction to Django</u> ➤ Define about Django ➤ Explain the Features of Django ➤ Define application of Django Framework ➤ Define the setup for Django Environment ➤ Explain Folder Structure <u>Session 2 – Model View Template Architecture</u> ➤ Define Model View Template (MVT) ➤ Illustrate the Structure of MVT ➤ Explain how does Django follow MVT ➤ List the differences between Model View Template (MVT) and Model View Controller (MVC)	Django Framework for Python SG – Session 1 & 2 XP – Session 1 & 2 TG - Session 1 & 2
2	DJANG – TL2	All the topics as listed below from Session 3 of <i>Django Framework for Python</i> book should be covered in this session. <u>Session 3 – Creating a Website</u> ➤ Explain how to start a project ➤ Explain how to create an app inside a project ➤ Explain how to activate app ➤ Explain how to create Templates, Views, URLs, and Models	Django Framework for Python SG – Session 3 XP – Session 3 TG - Session 3
3	DJANG – TL3	The Try It Yourself Questions from Session 1 to Session 3 of <i>Django Framework for Python</i> book should be covered in this session.	Django Framework for Python (Onlinevarsity) Session 1-3
4	DJANG – TL4	All the topics as listed below from Session 4 of <i>Django Framework for Python</i> book should be covered in this session.	Django Framework for Python SG – Session 4 XP – Session 4

Session No.	Session Title	Session Details	Deliverables' Mapping
		<u>Session 4 – Admin Panel and Admin Interface</u> ➤ Explain Django Admin Interface ➤ Explain Admin Interface ➤ Elaborate on how to create <i>superuser</i> ➤ Elaborate customizing admin panel	TG - Session 4
5	DJANG – TL5	All the topics as listed below from Session 5 of <i>Django Framework for Python</i> book should be covered in this session. <u>Session 5 – Django View Creation</u> ➤ Define and create views ➤ Identify types of views ➤ List the difference between Function-based and Class-based views ➤ Explain pros and cons of both Function-based and Class-based views ➤ Explain how to create Function-based type in views.py ➤ Describe operation of Class-based view ➤ Explain how to configure URL for different kind of views	Django Framework for Python SG – Session 5 XP – Session 5 TG – Session 5
6	DJANG – TL6	The Try It Yourself Questions from Session 4 and Session 5 of <i>Django Framework for Python</i> book should be covered in this session.	Django Framework for Python (Onlinevarsity) Session 4 & 5
7	DJANG – TL7	All the topics as listed below from Session 6 of <i>Django Framework for Python</i> book should be covered in this session. <u>Session 6 – Django Forms Creation</u> ➤ Define Django forms ➤ Explain HTTP, GET, and POST Method ➤ Explain how to create Connect Model or Model Fields with form ➤ Explain Instantiate, process, and render a simple form ➤ Explain how to implement different fields in form ➤ Elaborate how to develop form field validation ➤ Explain Set Widgets for Form Fields ➤ Explain and build arguments for the form	Django Framework for Python SG – Session 6 XP – Session 6 TG – Session 6

Session No.	Session Title	Session Details	Deliverables' Mapping
8	DJANG – TL8	All the topics as listed below from Session 7 of <i>Django Framework for Python</i> book should be covered in this session. <u>Session 7 – Django Templates</u> ➤ Explain Django templates ➤ Explain how to design Template folders ➤ Explain how to create HTML files inside Template folder ➤ List the conditions in Template ➤ Outline Tag in Template ➤ Define Form in Template ➤ Explain Rendering context in template ➤ Outline Template Filters	Django Framework for Python SG – Session 7 XP – Session 7 TG – Session 7
9	DJANG – TL9	The Try It Yourself Questions from Session 6 and Session 7 of <i>Django Framework for Python</i> book should be covered in this session.	Django Framework for Python (Onlinevarsity) Session 6 & 7
10	DJANG – TL10	All the topics as listed below from Session 8 of <i>Django Framework for Python</i> book should be covered in this session. <u>Session 8 – Django Serializers</u> ➤ Explain Django Serializer and Deserializer ➤ Explain how to install Serializer ➤ Explain how to create and Import Serializer ➤ List different types of Serializers ➤ Define Fields and Arguments for Serializer ➤ Describe View Set and URLs with Serializer	Django Framework for Python SG – Session 8 XP – Session 8 TG – Session 8
11	DJANG – TL11	All the topics as listed below from Session 9 of <i>Django Framework for Python</i> book should be covered in this session. <u>Session 9 – Django's REST Framework</u> ➤ Outline the Installation process ➤ Illustrate rest_framework to INSTALLED_APPS ➤ Define and Explain Serialization ➤ Define a ViewSet ➤ Explain URLs of API ➤ Explain Run Server and Check API	Django Framework for Python SG – Session 9 XP – Session 9 TG – Session 9

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Define and Explain CRUD Operations	
12	DJANG – TL12	The Try It Yourself Questions from Session 8 and Session 9 of <i>Django Framework for Python</i> book should be covered in this session.	Django Framework for Python (Onlinevarsity) Session 8 & 9

6. Library References

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| ➤ Django for Beginners: Build Websites with Python and Django by William S Vincent |
| ➤ Django 4 By Example: Build powerful and reliable Python web applications from scratch by Antonio Mele and Bob Belderbos |
| ➤ Django for Professionals: Production websites with Python & Django by William S Vincent |

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